

End-to-End Sales Analysis

Capstone Project — Week 4: Data Visualization & Storytelling
Data Analyst Course | Beeskilled — AICTE Virtual Internship

Executive Summary

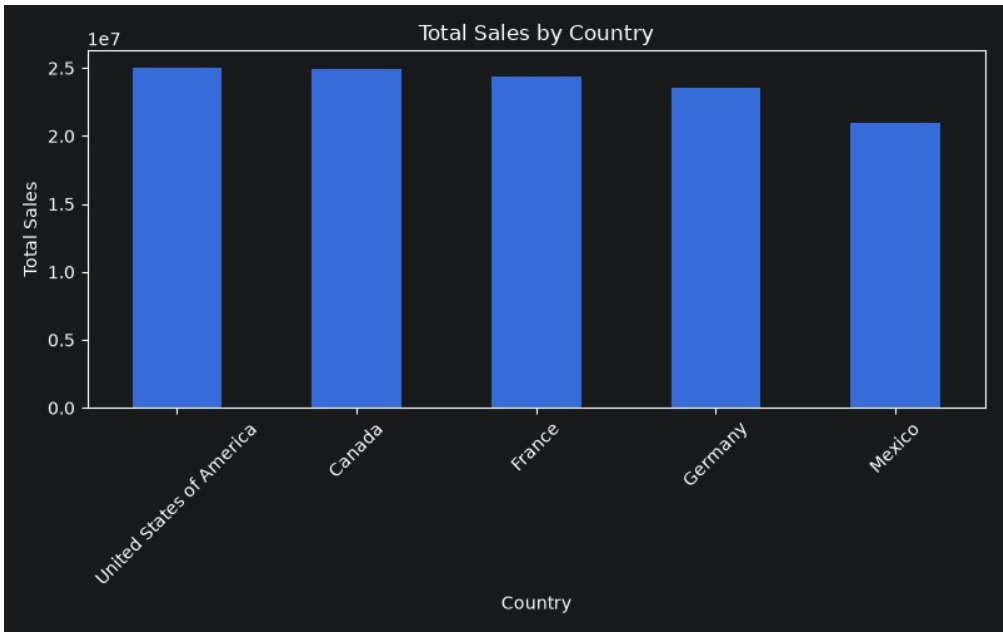
This report presents an end-to-end analysis of company sales data covering 700 transactions across five countries, five customer segments, and six products between 2013 and 2014. The dataset was explored using Python for EDA, SQL for business queries, and a Linear Regression model to predict sales. The findings below translate the results into clear, actionable recommendations for stakeholders.

\$118.7M Total Sales	\$16.9M Total Profit	14.2% Avg. Margin	700 Records Analyzed
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1. Exploratory Data Analysis (EDA)

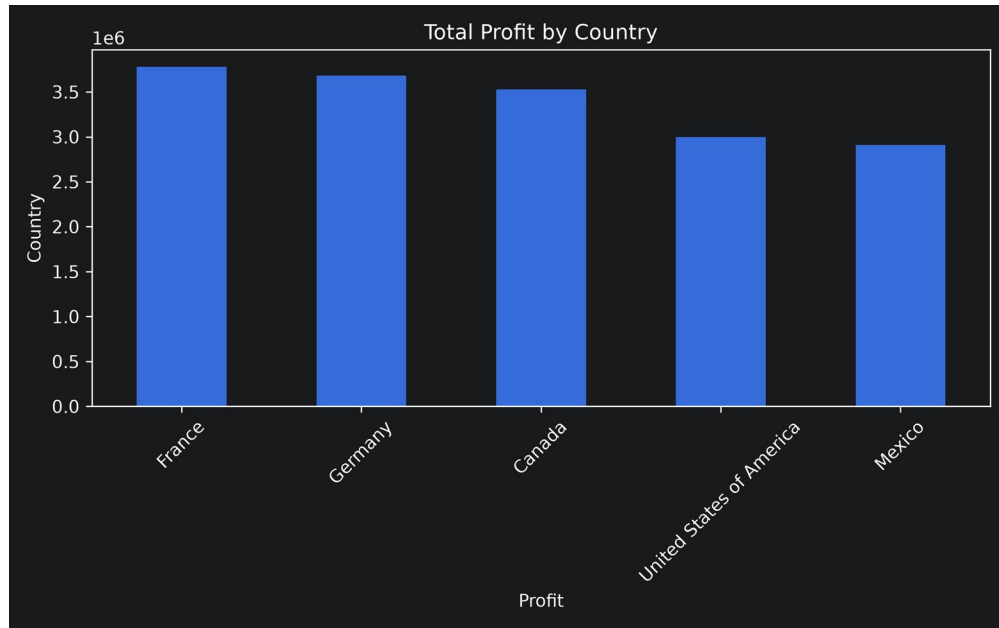
Total Sales by Country

The United States and Canada lead in total sales, each contributing roughly \$25M, followed closely by France and Germany. Mexico trails with the lowest total sales of the five markets.



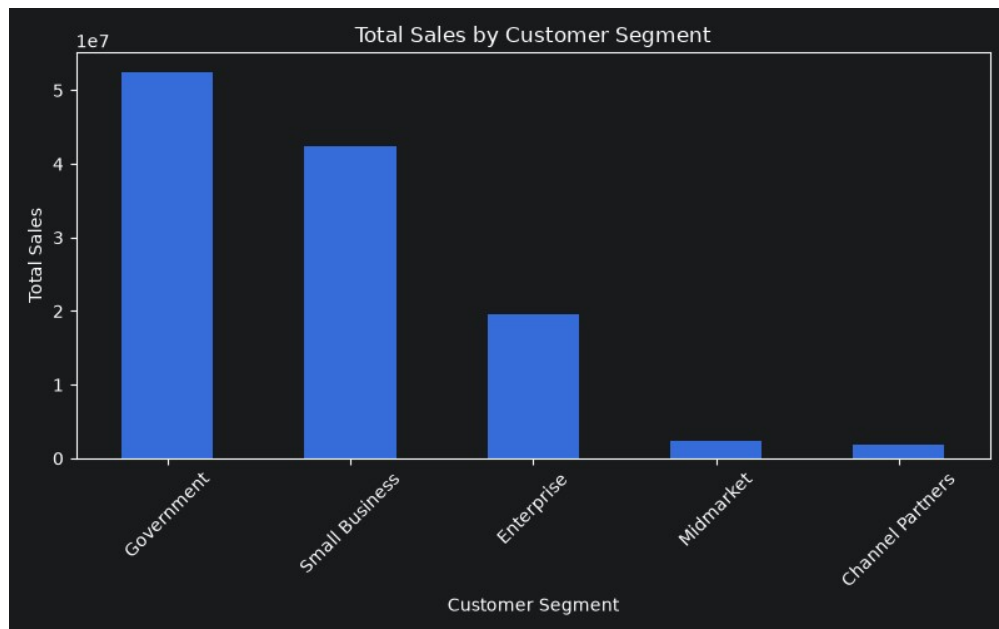
Total Profit by Country

Despite the US leading in sales volume, France and Germany generate the highest total profit — a sign that these markets convert revenue into profit more efficiently, likely due to lower average discounting.



Total Sales by Customer Segment

Government is by far the largest segment, generating over \$52M in sales, followed by Small Business (~\$42M) and Enterprise (~\$19.5M). Midmarket and Channel Partners are minor contributors by comparison.



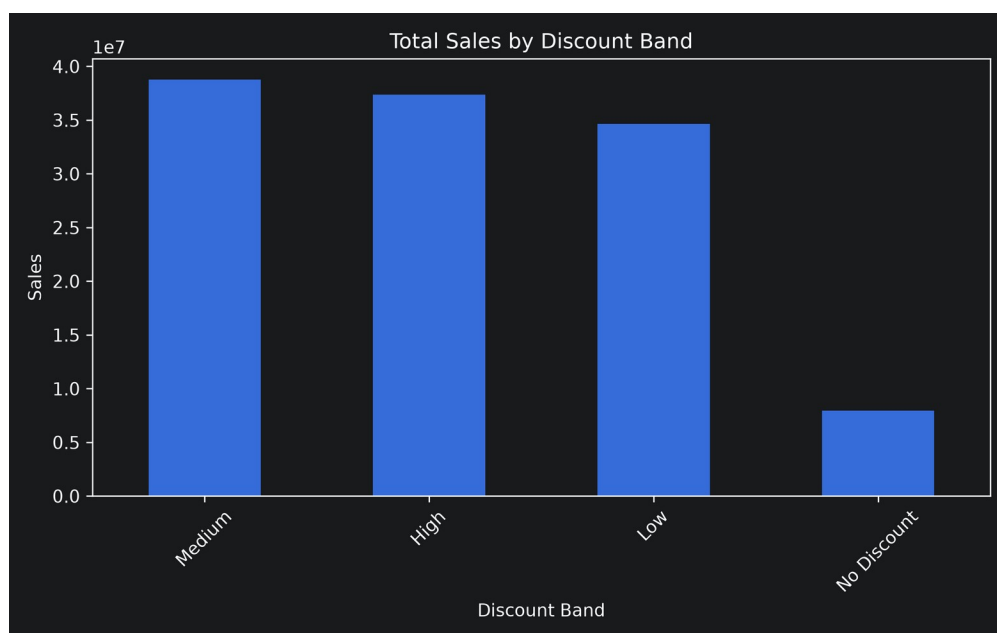
Total Profit by Product

Paseo is the standout product, generating close to \$4.8M in profit — well ahead of VTT and Amarilla. Carretera and Montana contribute the least profit among the six products.



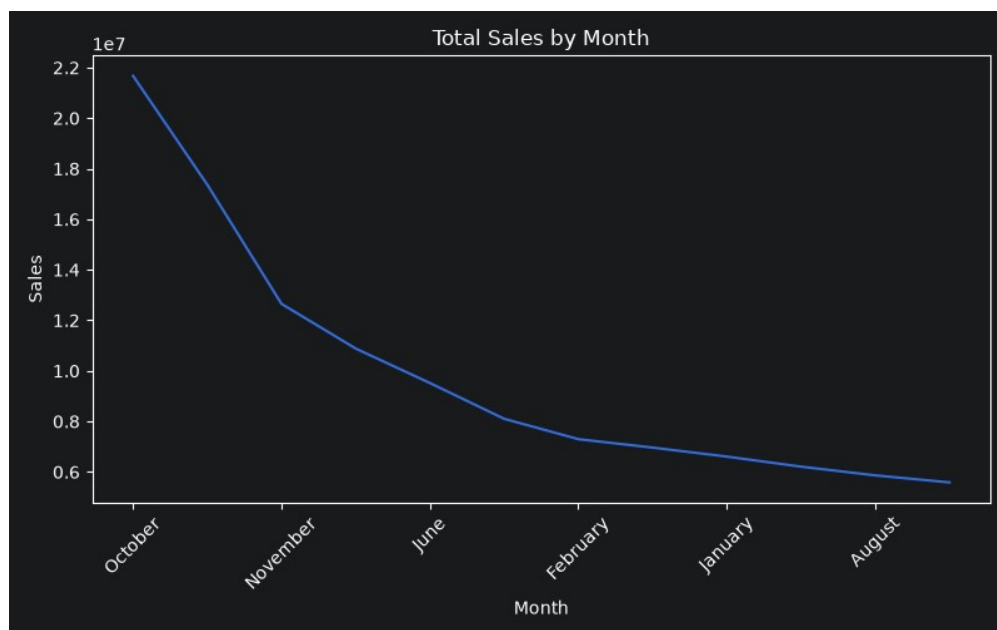
Total Sales by Discount Band

Sales under the Medium and High discount bands are the largest by volume (~\$38.7M and ~\$37.3M respectively), while transactions with no discount applied make up a much smaller share (~\$7.9M) of total sales — though, as the SQL analysis below shows, they carry the healthiest margins.



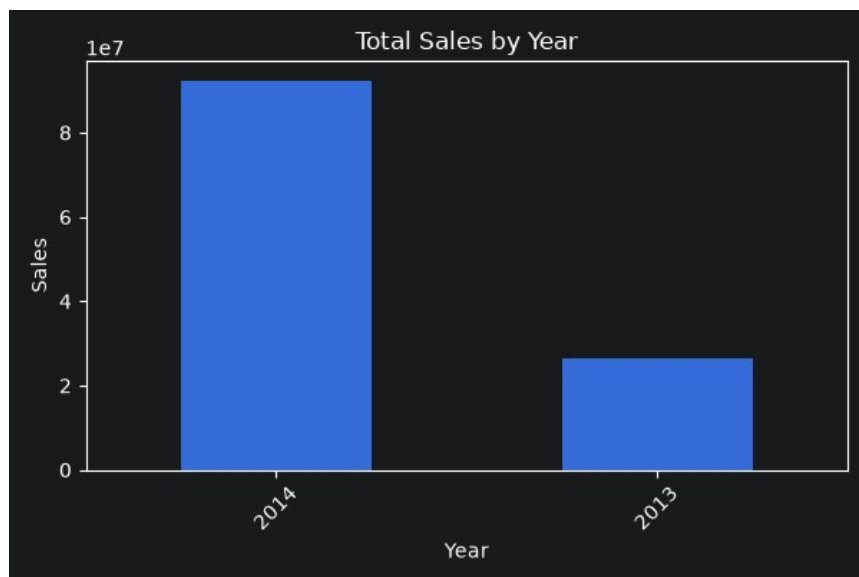
Total Sales by Month

October stands out as the strongest month by a wide margin (~\$21.7M), more than 1.5x the next highest month. Sales taper off through the rest of the year, with August the weakest of the months shown.



Total Sales by Year

2014 sales (~\$92M) dwarf 2013 sales (~\$26M) — though this partly reflects that the dataset covers a fuller 12 months of 2014 versus a partial year of 2013.



2. SQL Business Analysis

The following business questions were answered directly against the sales table using SQL:

Query Set

```
-- Total sales, total profit, profit margin
SELECT SUM(sales) AS total_sales FROM sales;
SELECT SUM(profit) AS total_profit FROM sales;
SELECT ROUND((SUM(Profit)/SUM(Sales))*100,2) AS Profit_Margin FROM sales;
```

```
-- Highest sales by country, segment, product, month, year
SELECT Country, SUM(Sales) AS total_sales FROM sales
GROUP BY Country ORDER BY total_sales DESC;

-- Top 10 highest individual sales transactions
SELECT Country, Product, Segment, Sales FROM sales
ORDER BY Sales DESC LIMIT 10;

-- Top 5 most profitable products
SELECT Product, SUM(Profit) AS Total_Profit FROM sales
GROUP BY Product ORDER BY Total_Profit DESC LIMIT 5;
```

(Full query file with all 13 queries — total sales/profit, breakdowns by country, segment, product, month, year, discount band, average profit by country, top 10 transactions, profit margin, top 5 profitable products, and average sales by segment — is included in the submission.)

Key Results

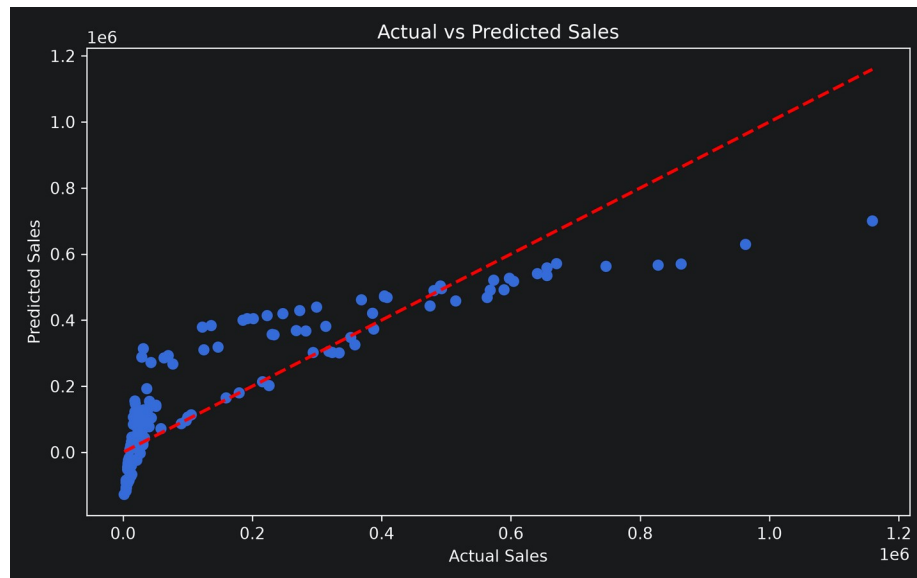
- Highest-selling country: United States (~\$25.0M), narrowly ahead of Canada (~\$24.9M).
- Highest-selling segment: Government (~\$52.3M) — more than the next two segments combined.
- Highest-selling product: Paseo — also the most profitable product overall.
- Best month for sales: October, well ahead of every other month.
- Highest-selling year: 2014, contributing roughly 3.5x the sales of 2013.
- Most profitable country on average: France, followed closely by Germany — despite the US having higher raw sales volume.
- Overall profit margin across all transactions: approximately 14.2%.

3. Regression Model — Predicting Sales

A Linear Regression model was trained to predict Sales using the available transaction features, then evaluated on a held-out test set:

Model	R ²	MAE (\$)	RMSE (\$)
Linear Regression	0.76	\$83,644	\$114,727

The model explains about 76% of the variation in Sales ($R^2 = 0.76$), with an average prediction error of roughly \$83,600. This is a solid baseline result for a simple linear model.



The Actual vs. Predicted scatter plot shows the model tracks well for low-to-mid range sales values, where most points sit close to the red diagonal (perfect prediction) line. At higher sales values, the model increasingly under-predicts — the points fall below the line — indicating Linear Regression struggles to capture the full scale of the largest transactions. This is a common limitation of linear models when the relationship between features and target isn't perfectly linear; a tree-based model (e.g. Random Forest) would likely reduce this gap.

4. Key Findings

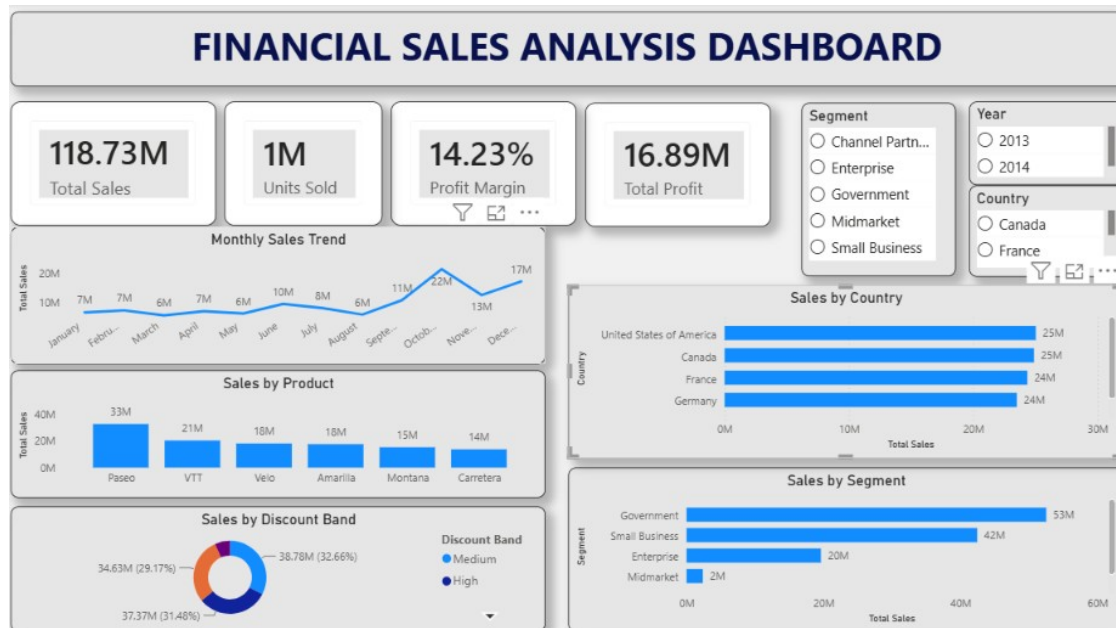
- Paseo is both the top-selling and most profitable product, making it the clearest priority for inventory and marketing focus.
- Government and Small Business are the two dominant customer segments — together driving over \$94M of the \$118.7M in total sales.
- The US leads in sales volume, but France and Germany lead in profit — discounting patterns in the US market are likely compressing margins there.
- October is a standout month for sales, suggesting a seasonal or promotional effect worth investigating further.
- The Linear Regression model ($R^2 = 0.76$) is a reasonable baseline for predicting sales, but under-predicts high-value transactions — a non-linear model would likely improve accuracy at the top end.

5. Recommendations

- Review discount policy in the US market specifically, since higher sales volume there isn't translating into proportionally higher profit.
- Prioritize Paseo in stock planning and sales campaigns given its outsized contribution to both revenue and profit.
- Investigate what drove October's spike (promotions, seasonality, contract renewals) to see if it can be replicated in other months.
- Continue investing in the Government and Small Business segments, as they are by far the largest revenue drivers.
- Consider testing a Random Forest or other non-linear model to better predict high-value transactions, where Linear Regression currently under-performs.

6. Interactive Power BI Dashboard

A one-page Power BI dashboard was built on the cleaned dataset to give stakeholders a self-service view of the same metrics covered in this report, with the ability to filter by Segment, Country, and Year.



Dashboard components:

- KPI cards — Total Sales (\$118.73M), Units Sold (1M), Profit Margin (14.23%), and Total Profit (\$16.89M), matching the figures reported above.
- Monthly Sales Trend — a line chart of sales by month, showing the same seasonal pattern identified in the EDA (a strong spike late in the year).
- Sales by Product — confirms Paseo as the clear leader (~\$33M), consistent with it also being the most profitable product.
- Sales by Discount Band — a donut chart breaking down the ~29-32% split across Medium, High, and Low discount bands.
- Sales by Country and Sales by Segment — bar charts mirroring the country and segment findings from the EDA, letting a stakeholder drill in directly.
- Segment, Year, and Country slicers — let a viewer filter the entire dashboard to a specific segment, year, or country in one click, without needing to touch the underlying data.

This turns the static findings in this report into something a stakeholder can explore on their own — for example, filtering to just "Government" and "2014" to check whether the October spike was concentrated in that segment.

7. Deliverables

- Cleaned dataset — ready for Power BI/Tableau import.
- SQL query file (sales_analysis.sql) with all business questions and answers.
- Python regression model code and evaluation metrics.
- EDA chart set (8 visualizations) shown above.
- Power BI dashboard (.pbix) with KPI cards, trend charts, and interactive filters.
- This stakeholder report.